

GAYATRI SHARMA KURMATEY

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PROFESSIONAL SUMMARY

Data Engineer and AI/ML Engineer with 6+ years of experience building ETL pipelines, BI dashboards, data models, automation systems, and machine learning workflows across supply chain, education, energy, automotive, logistics, and research domains. Skilled in SQL, Python, Azure Synapse, Power BI, Tableau, PostgreSQL, machine learning, deep learning, and Generative AI workflows. Proven record of reducing manual reporting time, improving KPI visibility, developing ML-enabled analytics solutions, and building executive dashboards for large-scale operational and CAPEX decision-making.

SKILLS

Automation & Workflow: Airflow, Power Apps, Power Automate, Power Pages, Dataverse, Kubernetes, Microsoft Copilot, ChatGPT, Cursor, Codex, Claude

BI & Data Visualization: Chimera, Power BI, Tableau, Qlik Sense, Google Analytics, Matplotlib, Seaborn, Visio

Data Formats: JSON, CSV, XLSX, SQL, PDF, GPickle, FASTA, PDB, image files (PNG, JPEG)

Data Science: BioPython, Docker, Jupyter Notebook, KarateClub, NetworkX, Python (Scikit-Learn, TensorFlow, PyTorch), R, SAS

Databases, Warehousing & ETL: PostgreSQL, Microsoft SQL Server, MySQL, MongoDB, Sybase, SQLdbx, BigQuery, Snowflake, Azure Data Factory, SSIS, dbt, ETL/ELT Pipelines, Data Modeling, Data Warehousing

Governance & Enterprise Systems: Data Governance, Access Control, RBAC, Data Privacy

Excel & Scripting: Advanced Excel (Macros, Pivot Tables, VBA, and Advanced Functions), Google Sheets

Experimentation & Analysis: A/B Testing, Case Study Testing, Experimental Design, Hypothesis Testing, Statistical Modeling

Financial Analytics: AML, Fraud Detection, KPI Modeling, Risk Dashboards, Transaction Monitoring, FP&A

Machine Learning, Deep Learning & Generative AI: LLMs, BERT, Transformers, Foundation Models, CNNs, Mask R-CNN, Graph Neural Networks, Geometric Deep Learning, Semantic Modeling, PyTorch, TensorFlow, Scikit-Learn, SageMaker

Project & Team Management: Confluence, JIRA, Microsoft Project, Monday.com

Reporting & Business Analytics: KPI Dashboards, Executive Reporting, Operational Dashboards, Weekly/Monthly Business Reporting, Data Storytelling, DAX, Power Query

Cloud & Infrastructure: Azure, Google Cloud, Microsoft Intune, SharePoint, Snowflake, AWS, AWS S3, S3 Buckets, Amazon SageMaker

Trust, Safety & Compliance: Access Control, Role-Based Security, Policy Enforcement, Risk Monitoring, Data Privacy

WORK EXPERIENCE

Belong Automotive Technologies, LLC, Texas, *Project Lead Data Engineer*

Oct 2025 – Present

- Lead 2 interns in building Odoo ERP workflows, Python ETL pipelines, and PostgreSQL data models across 50+ supply chain functions, improving operational visibility across HR, inventory, purchasing, supplier, and demand planning processes
- Partner with China and U.S. operations teams to translate inventory, purchasing, supplier, and demand planning requirements into scalable ERP workflows, data models, and reporting assets, improving global supply chain coordination and decision-making
- Reduced manual reporting effort by 12 hours/week and improved KPI visibility from 9 days to 2 hours through automated reporting workflows, saving 600+ analyst hours annually and an estimated \$40K–\$75K in productivity value
- Accelerated U.S. launch readiness by securing the company domain, designing, and deploying the official U.S. website, setting up Microsoft 365, and creating communication workflows for 100+ team members in 8 weeks

SkyTech Inc, New Jersey, *Data Analyst (Contractor)*

Feb 2025 – Oct 2025

- Tested HR workflow automation configurations, improving dashboard adoption by 25%
- Integrated HR data across MSSQL, APIs, Tableau, and Power BI, improving data interoperability by 30% and enabling faster workforce reporting across compliance, HR operations, and leadership dashboards
- Automated RBAC and compliance workflows in Azure Synapse, strengthening governance controls, reducing manual access-review effort by 15+ hours/week, and lowering operational risk across HR data processes

Robert Half, Fremont Unified School District, California, *Data Science Engineer (Contractor)*

Mar 2024 – Jan 2025

(via SkyTech Inc from Oct 2024)

- Led a 6-member team to build SQL, Azure Synapse, and Power Platform ETL pipelines for student database migration, saving 25 hours/week in manual processing and improving reliability of district-level reporting operations
- Built Power BI access-control dashboards for IT teams to track faculty laptop activations and inactive users, reducing activation follow-up time by 15 hours/week
- Automated inclusion, diversity, and workforce equity reporting for 3,360+ students, improving student data validation accuracy by 30%
- Streamlined ERP-integrated business process automation for 2,500+ asset and HR workflows, improving KPI tracking, forecasting accuracy, and data reliability across finance, IT, and workforce operations
- Coached cross-functional teams on Power Platform and Tableau dashboards, driving 100% adoption within one week and saving 18 hours/week in recurring manual reporting effort

Pacific Gas and Electric Company, California, *Business Intelligence Analyst (Contractor)*

May 2023 – Aug 2023

- Optimized Spark SQL data lake pipelines for operational dashboards, reducing refresh time from 60 minutes to 20 minutes and improving timely access to business-critical metrics for utility operations teams
- Automated SQL-to-SharePoint reporting workflows using Python, Tableau, and Power BI, saving 20+ hours/week and reducing recurring manual reporting costs for operational analytics teams
- Trained analysts on Power Query, DAX, and Tableau calculations, reducing recurring dashboard support requests by 30% and saving 10+ hours/week

Tesla Inc, California, Data Science Engineer Intern

May 2022 – Dec 2022

- Developed space planning and people analytics dashboards using Power Platform and Dynamics 365, saving 10+ hours weekly
- Created 20+ Power BI and Tableau dashboards for PMs and executives monitoring \$1.3B+ CAPEX projects, improving portfolio visibility and reducing report refresh time by 40 minutes daily through DAX optimization
- Built S-curve forecasting tool for 500+ construction projects, applying statistical testing and predictive analytics to support \$1.3B+ CAPEX planning, improve logistics and financial forecasting, and save 20+ hours/week
- Enforced role-based access controls via Azure AD and SharePoint, strengthening data governance and ensuring compliance across Power Platform systems

Tango Analytics, Texas, Data Scientist Intern

Jan 2022 – May 2022

- Engineered deep learning computer vision pipelines using OpenCV and Mask R-CNN to detect and annotate floor plan objects, improving dataset quality for object recognition and downstream ML workflows
- Cleaned, labeled, and validated image datasets for entity recognition models, and semantic segmentation, achieving 88% accuracy and reducing production integration effort by 65%

Amazon, India, Associate Logistics Data Analyst

Aug 2018 – Aug 2019

- Improved last-mile logistics analytics by uncovering customer delivery insights, boosting delivery efficiency by 80%, resolving 100+ operational issues, and saving 110+ hours/month across logistics support workflows
- Analyzed Amazon Flex pilot feedback and delivery performance data to improve onboarding, training, software workflows, and escalation resolution across U.S.–India logistics operations

Indian Space Research Organization, India, Data Science Intern

May 2017 – Mar 2018

- Built a materialized-view analytics system for procurement data using multidimensional data modeling, Sybase data extraction, MySQL queries in SQLdbx, and COWAA interface integration
- Developed an HTML, CSS, and Java-based user interface to visualize query results and support procurement analysis for business decision-making
- Developed a conference hall booking system using Sybase, MySQL, HTML, CSS, and Java, enabling ISRO employees to manage room reservations, prevent scheduling conflicts, and improve booking process efficiency by 99%

RESEARCH EXPERIENCE**NASA Ames Research Center, California, Open-Source Researcher**

Feb 2024 – Present

- Tested graph-based machine learning approaches for protein structure clustering by converting protein structures into atomic-level graph representations using BioPython and NetworkX
- Evaluated Graph2Vec embeddings with Weisfeiler-Lehman labeling strategies to analyze structural patterns across active and inactive kinase conformational states
- Applied PCA, t-SNE, and KMeans clustering to visualize embedding separability and assess limitations of topology-driven graph representations
- Explored geometry-aware deep learning methods, including geometric GNNs, SE(3)-equivariant transformers, and SO(3)-equivariant autoencoders, as potential approaches for improving protein structure representation learning

Jet Propulsion Laboratory, NASA, California, Volunteer Research Assistant

May 2023 – Aug 2023

- Compiled Generative AI, LLMs, BERT, and Haystack for semantic modeling, knowledge extraction, and advanced R&D analytics

The University of Texas at Dallas, Research Assistant

Jan 2022 – May 2022

- Coded containerized Open Data Cube for NASA, automating 500+ satellite image ingestions via Docker Containers and AWS S3 Bucket for scalable storage

The University of Texas at Dallas, Machine Learning Research Assistant

Aug 2021 – Dec 2021

- Tested ML models (SVM, Neural Networks, Logistic Regression, and Stacking) with feature selection to predict road accident severity

EDUCATION**Master of Science - Information Technology & Management, University of Texas at Dallas**

Jan 2021 – Dec 2022

Bachelor of Technology - Computer Science, Jawaharlal Nehru Technological University

Sep 2014 – Jun 2018

HONORS & AWARDS

- Recognized by superintendents, district officials and Vice Presidents for automation and dashboard initiatives at Tesla and FUSD
- Scholar with Recognition – Ranked Top 3 GPA in MS-ITM Class, University of Texas at Dallas

CERTIFICATES

Semantic Segmentation with Amazon SageMaker with Amazon AWS	May 2026
Introduction to Large Language Models (LLM)	May 2026
Introduction to Generative AI	May 2026
Applying AI Principles with Google Cloud	May 2026
Databricks Generative AI Foundation	Jun 2026
IACET OSHA 30	Dec 2025
LinkedIn Certification in Kubernetes	Jul 2025
LinkedIn Certification in Advance Window Functions	Oct 2024
LinkedIn Certification in Tableau for Data Scientists	Nov 2021